Field	Entry
Source ID	WBC-0009
Title	Effect of Grain Angle on the Strain Distribution During Orthogonal Cutting of Hinoki (Chamaecyparis obtusa) Measured Using a Digital Image Correlation Method
Source type	Paper (open access)
Link / File	https://jwoodscience.springeropen.com/articles/10.1186/s10086-019-1824-2
Craft applicability	Wood
Primary Why-theme	10. Failure, Repair and Quality
Process stage	Primary shaping / rough carving
Questions answered / generated	Why does cutting against or across the grain produce torn or fuzzy grain? Why does grain angle affect surface tear-out during carving?
Evidence level	E3 (direct scientific evidence, strain measurement via digital image correlation)
Relevance note	Directly measures how grain angle relative to the tool path governs strain and cleavage ahead of the cutting edge, providing a mechanical explanation for the torn-grain and fuzzy-grain defects recorded during primary shaping and carving of Udayagiri wood.